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3.2.1. Numpy: Matrix Operations

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The given code takes two  matrices, , and , as input from the user and converts them into NumPy arrays.

**Task:**

You are required to compute and display the results of the following matrix operations:

1. **Addition** ()
2. **Subtraction** ()
3. **Element-wise Multiplication** ()
4. **Matrix Multiplication** ()
5. **Transpose of Matrix A**

**Input Format:**

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

**Output Format:**

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

```
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])



# Addition
print("Addition (A + B):")
print(matrix_a + matrix_b)

# Subtraction
print("Subtraction (A - B):")
print(matrix_a - matrix_b)

# Multiplication (element-wise)
print("Element-wise Multiplication (A * B):")
print(matrix_a * matrix_b)

# Matrix multiplication (dot product)
print("A dot B:")
print(np.dot(matrix_a, matrix_b))

# Transpose
```

```
print("Transpose of A:")

print(matrix_a.T)
```